

# Raphael Bernardo

Senior Android Engineer • *Ex-Uber*

Architect of a streaming framework powering  
50+ TV brands — native (JNI/C++), TV & multi-  
DRM expertise.

LOC Rio de Janeiro • GMT-3

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## § 01 Summary

PROFILE • POSITIONING

**Senior Android engineer with cross-stack reach — Uber Maps SDK (JNI/C++), OTT / CTV streaming frameworks, broadcast-grade DRM.** Ten-plus years shipping production apps across Android, Android TV, Fire OS, webOS and Tizen — including an OTT streaming framework powering 50+ broadcast brands and 4K multi-DRM playback. As an independent contractor I run multiple client engagements in parallel — at any given time only one is full-time, with the rest as part-time concurrent contracts.

## § 02 Experience

ENGAGEMENTS • 2014 → PRESENT

### Nexstar Media Group, Inc.

Aug 2025 — Present • current

Senior Fire OS / Android TV Engineer • via Nimble.LA • full remote

FULL-TIME CONTRACT

- Architected an enterprise **OTT streaming framework** powering **50+ TV station brands** across Fire TV and Android TV using **MVVM + Clean Architecture** with a plugin-based integration layer for third-party services.
- Built five specialised **Media3 / ExoPlayer** implementations covering VOD, live, and audio playback with HLS adaptive bitrate.
- Shipped DRM-protected playback (**Widevine, PlayReady, FairPlay**) with both server-side and client-side ad insertion.
- Integrated **12+ analytics platforms** (Adobe Analytics, mParticle, Nielsen DCR, ComScore, Permutive, BlueConic) under GDPR / CCPA compliance.
- Delivered TV-optimised authentication: device-based passwordless, magic link, and QR-code login; entitlement checking and flexible paywall integration.
- Enabled zero-downtime feature rollouts across **50+ product flavors** via remote configuration; built TV UI on Android Leanback with full d-pad navigation.
- Refactored callback chains to reactive Kotlin **SharedFlow**; resolved background-audio lifecycle issues and lifecycle-aware leaks in fragment navigation.
- Wired **Firebase Crashlytics + New Relic** for crash and performance monitoring; coverage with JUnit, Mockk, Robolectric, Espresso.

Kotlin   Coroutines / Flow   Media3 / ExoPlayer   Dagger 2   Retrofit2   Moshi   Glide  
Android Leanback   Jetpack (Nav, ViewModel, Room)   Firebase   Google IMA   Adobe Marketing SDK  
mParticle   Nielsen   ComScore   New Relic   Mux Analytics   JUnit · Mockk · Espresso

## FreeCast, Inc.

Mar 2021 — Jul 2025 · 4 yrs 5 mo

Android / Android TV / Fire OS Engineer · full remote   LONG-TERM PART-TIME CONTRACT

- Led the **full migration of the legacy Java codebase to Kotlin**; introduced Use-Case + Clean Architecture and improved end-user UX.
- Implemented **HLS / DASH** playback for **live linear TV and VOD** with Widevine DRM-protected streaming; integrated Google Cast and a paid subscription gateway.
- Maintained and developed companion apps on **webOS and Tizen** alongside the Android TV / Fire OS surfaces.
- Integrated **OTA tuner platforms** (HDHomeRun, AirTV, Alticast, PDAQ) with GStreamer-based media pipelines for secure live broadcast over HLS / DASH.
- Owned **DRM, EPG, tuner orchestration, transcoding**, and multi-device streaming end-to-end.
- Shipped on Google Play (*Select TV Android, FreeCast Watch*) and Amazon (*FreeCast Watch Fire OS*).

Kotlin   Jetpack Compose   Hilt   Coroutines / Flow   Media3   Room · DataStore   Retrofit2  
Picasso   Firebase Analytics · Crashlytics · FCM   Mixpanel   webOS · Tizen  
HDHomeRun · AirTV · Alticast · PDAQ   GStreamer   Widevine DRM

## Claro TV+

Apr 2024 — Dec 2024 · 9 mo

Android TV / Fire OS Engineer · full remote   CONCURRENT PART-TIME CONTRACT

- Greenfield Android TV / Fire OS app for one of Brazil's largest pay-TV operators; **4K streaming** via DASH / HLS with multi-DRM.
- Architected the data layer as a shared **Kotlin Multiplatform (KMP)** submodule consumed by Android, webOS and Tizen clients.
- Multi-module **MVVM** architecture; Media3 playback; analytics across Firebase + Google Analytics 4.

Kotlin Multiplatform (KMP)   Kotlin   Jetpack Navigation · ViewModel · Room   Hilt   Media3  
Coroutines / Flow   Firebase + GA4   webOS · Tizen

## Uber Technologies

May 2021 — May 2022 · 1 yr

Android Engineer · Maps SDK team · via Matchpoint Solutions · full remote

CONCURRENT PART-TIME CONTRACT

- Worked on the **Maps team SDK** module — JNI/C++ bindings between Uber's native Maps engine and the Android client.
- Contributed to Uber's open-source Android architecture framework (**RIBs**); shipped to Uber Rider, Uber Eats and Uber Driver.
- Operated inside a 10K+ engineer codebase using **Buck, Flipper**, custom analytics / logger / crash reporter and a feature-by-module architecture.

Java · Kotlin   C++ · NDK · JNI   Groovy   Python · SQL   RIBs   Buck   CMake   Flipper  
Firebase

## Invillia

Apr 2020 — Aug 2020 · 5 mo

Android Engineer · full remote   PART-TIME CONTRACT

- Re-modelled the home page of **Casas Bahia** and **Pontofrio** — converted a static product shelf into a fully dynamic, backend-driven shelf system.
- Implemented end-to-end analytics tracking for product clicks and shelf scroll-depth via Firebase Analytics and custom API calls.

Kotlin   MVVM   Dagger 2   LiveData   Retrofit2   Picasso



Firebase (Analytics · FCM · Remote Config · Crashlytics)

Jenkins CI/CD

## Super Revendedores

Oct 2018 – Feb 2021 · 2 yrs 5 mo

Android Engineer · full remote

- Migrated the full codebase from Java to Kotlin; implemented **online payment gateway**, virtual showcases and an SSO link to a partner web ordering portal.
- Built scalable asynchronous order processing on **AWS SQS / SNS** for reliable queuing and notification delivery.

Kotlin

AWS (SQS · SNS · Storage)

Firebase (Firestore · Auth · Remote Config · IAM · FCM)

Facebook OAuth

DbFlow

Dagger 2

MVP

Retrofit2

## Concrete · an Accenture Company

Dec 2016 – Oct 2018 · 1 yr 11 mo

Jr. Android Engineer

- Built the **Minha CVC** travel app from scratch — **1M+ downloads**; shop geolocation directing customers to physical stores; itinerary feature reducing customer-service call volume.
- Worked on **Getnet**; migrated MVP to MVVM; introduced RxJava2 and LiveData.

Java 8

Kotlin

RxJava2 · LiveData

MVP → MVVM

Dagger 2

Google Maps API

Retrofit2 · Picasso

Firebase

Jenkins CI/CD

## PraPagar (fintech)

Jun 2015 – Dec 2016 · 1 yr 7 mo

Jr. Android Engineer · collaborating with US team (MaxxPotential)

- Built mobile payments via **QR code and geolocation**; first Android + Laravel REST experience.

Android

PHP · Laravel 5

REST

OOP · Design Patterns

## FASOLTI · Fábrica de Soluções em TI

Sep 2014 – Jan 2015 · 5 mo

Test Developer

INTERNSHIP

- Documented and implemented automated test cases in **Selenium with TestNG**.

## IBM Brasil

Jun 2014 – Dec 2014 · 6 mo

Software Engineering Intern · Rio de Janeiro · while pursuing B.S. at Instituto Infnet

- Wrote **SQL queries against IBM DB2** in support of contract and sales teams; produced reporting spreadsheets and Hyperion-based BI artefacts.
- Documented internal processes via SharePoint; first professional exposure to enterprise databases, REST APIs, and Agile workflows.

IBM DB2

SQL

IBM Hyperion

SharePoint

Python

REST APIs

Git

# Raphael Bernardo

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## § 03 Selected Personal Work

INDEPENDENT · OPEN SOURCE · RESEARCH

### iSpeakPDF

Privacy-by-design web platform converting PDFs to natural-sounding audio entirely in-browser via WebAssembly + Web Workers — no document leaves the device. WCAG-aligned UI, on-device neural TTS with sentence chunking, OCR fallback for scans, subscription tiers and usage limits.

Next.js · React

TypeScript

WebAssembly · Web Workers

PDF.js

WebGPU

Supabase

Stripe

### Bitcoin Keyspace Explorer

Educational web app for navigating the Bitcoin private-keyspace — BigInt page jumps, P2PKH / P2WPKH / P2SH-P2WPKH / P2TR derivations, real-time balance queries against a Postgres cache, virtualised lists, background scanning sessions with Telegram notifications.

Next.js 14

TypeScript

PostgreSQL

Node.js APIs

MUI

### Beer Price Calculator

Android app to find the best beer value (price / ml) with history, mini line / bar / pie / heat-map insights, PT / EN localisation, Play Billing subscriptions and coupon-based unlocks. Published on the Play Store.

Kotlin

Jetpack Compose

Hilt

Room

Coroutines / Flow

Play Billing

Material 3

### Custom BTC Cache Miner

Research-grade multi-pool Bitcoin SHA-256d **CUDA** miner that tests the same nonce across concurrent Stratum and GBT sources with safe hot-swaps and a VRAM cache (~85% utilisation, dedup, adaptive watermarks). CPU double-SHA-256 verify, crash-safe outbox.

C++ · CUDA

CMake

nlohmann::json

OpenSSL

Windows · Linux

## § 04 Education

FORMAL · CONTINUING

### Instituto Infnet · Rio de Janeiro

2013 — 2018

B.S. Computer Engineering — software development, mathematical modelling, OS & hardware architecture.

### Chiswick House School (Malta)

1997 — 2005

K-12 / High School Diploma.

### Orion Consulting · Rio de Janeiro

2010

Java fundamentals — OOP, software design, code patterns and logic.

## § 05 Technical Stack

REFERENCE · ATS INDEX

### LANGUAGES

Kotlin

Java

C++

TypeScript

JavaScript

Python

Groovy

SQL

PHP (Laravel)

### MOBILE / TV

Android SDK

Android TV

Fire OS

webOS

Tizen

Android Leanback



STREAMING ·  
MEDIA

- Jetpack Compose
- Compose for TV
- Kotlin Multiplatform (KMP)
- NDK · JNI
- OTT
- CTV (Connected TV)
- Linear TV
- VOD
- Live Streaming
- Media3 / ExoPlayer
- MediaCodec
- HLS
- LL-HLS
- DASH
- CMAF
- H.264 / AVC
- HEVC / H.265
- HDR10+
- AAC
- Hardware-accelerated decoding
- Widevine
- PlayReady
- FairPlay
- Google IMA SDK
- SSAI · CSAI
- GStreamer
- HDHomeRun · AirTV · Alticast · PDAQ

ARCHITECTURE

- MVVM
- MVI
- Unidirectional Data Flow
- Clean Architecture
- Use Case
- Multi-module
- RIBs
- MVP
- Plugin-based integration

JETPACK ·  
REACTIVE

- Coroutines · Flow
- SharedFlow · StateFlow
- LiveData
- ViewModel · Navigation
- Room · DataStore
- Hilt · Dagger 2
- RxJava2

CLOUD · BACKEND

- Firebase (Analytics · Crashlytics · FCM · Auth · Firestore · Remote Config · IAM)
- AWS (SQS · SNS · Storage)
- Supabase
- PostgreSQL
- REST APIs
- Retrofit2 · Moshi

ANALYTICS

- Adobe Analytics
- Adobe Marketing SDK
- mParticle
- Nielsen DCR
- ComScore
- Permutive
- BlueConic
- New Relic
- Mux
- Mixpanel
- Google Analytics 4

TESTING · BUILD

- JUnit
- Mockk
- Robolectric
- Espresso
- Paparazzi (screenshot)
- Turbine (Flow testing)
- Selenium · TestNG
- Buck
- Gradle
- CMake
- Jenkins CI/CD
- Flipper

PERFORMANCE ·  
OBSERVABILITY

- Quality of Experience (QoE)
- Time-to-First-Frame (TTFF)
- Rebuffering rate
- Startup-time optimisation
- SysTrace
- Perfetto
- Android Profiler

COMPLIANCE ·  
ACCESSIBILITY

- GDPR
- CCPA
- WCAG 2.2
- TalkBack · d-pad focus order
- DRM & Content Security
- OAuth · SSO

LANGUAGES  
(SPOKEN)

- Portuguese (native)
- English (fluent)